SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

TIRUCHIRAPPALLI CAMPUS

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COMPUTER NETWORKS LAB REPORT

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EXPERIMENT 1: Introduction to Packet Tracer & Cable Color Codes

Date: 23-10-2026

AIM

To study Cisco Packet Tracer, connect two PCs through a switch and check the communication between them using ping, and to study the straight-through and crossover cables and their color codes.

SOFTWARE USED

Cisco Packet Tracer

THEORY

Cisco Packet Tracer is a network simulation software made by Cisco. It lets us create networks, configure devices and test them without using real hardware. In peer-to-peer communication, computers talk to each other directly without requiring a dedicated server. A switch connects devices within the same local network and forwards data using MAC addresses.

Ethernet cables consist of 8 color-coded copper wires (4 twisted pairs) terminated with RJ-45 connectors. Two main wiring standards exist: **T568A** and **T568B**.

Wire Color Codes

Standard	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8
T568A	White-Green	Green	White-Orange	Blue	White-Blue	Orange	White-Brown	Brown
T568B	White-Orange	Orange	White-Green	Blue	White-Blue	Green	White-Brown	Brown

Cable Types & Applications

Cable Type	Wiring Scheme	Used to Connect	Examples
Straight-through	Same standard on both ends (T568B to T568B)	Different devices	PC to Switch, Switch to Router
Crossover	T568A on one end, T568B on the other	Similar devices	PC to PC, Switch to Switch

IP Addressing Scheme

Device Name	IP Address	Subnet Mask	Switch Port
PC0	192.168.1.1	255.255.255.0	Fa0/1
PC1	192.168.1.2	255.255.255.0	Fa0/2

PROCEDURE

1. Open Cisco Packet Tracer.
2. Select two PCs (PC0 and PC1) from *End Devices* and one 2960 Switch from *Switches*, placing them into the workspace.
3. Select a **Copper Straight-Through cable** and connect PC0 (FastEthernet0) to Fa0/1 of the switch, and PC1 (FastEthernet0) to Fa0/2 of the switch.
4. Configure PC0: Navigate to `Desktop` → `IP Configuration` and assign IP address `192.168.1.1` with subnet mask `255.255.255.0`.
5. Configure PC1: Navigate to `Desktop` → `IP Configuration` and assign IP address `192.168.1.2` with subnet mask `255.255.255.0`.
6. Wait until link status indicator LEDs on the switch turn from amber to green.
7. Open PC0 → `Desktop` → `Command Prompt` and type command: `ping 192.168.1.2`.
8. Observe and record the ICMP echo replies.

NETWORK TOPOLOGY & OUTPUT VERIFICATION

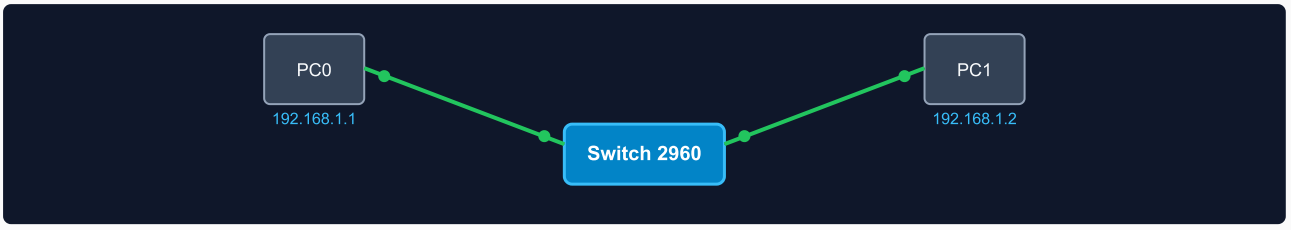


Fig 1.1 – Topology: PC0 and PC1 connected through a 2960 switch

```
PC0> ping 192.168.1.2 Pinging 192.168.1.2 with 32 bytes of data: Reply from 192.168.1.2: bytes=
```

Fig 1.2 – Command Prompt: Successful ping from PC0 to PC1 (192.168.1.2)

RESULT

Thus, two PCs were successfully connected using a 2960 switch in Cisco Packet Tracer, and end-to-end communication was verified via ping (4 packets sent, 4 received, 0% loss). The color code standards and cable configurations were studied.

EXPERIMENT 2: Router Configuration (Passwords & Interfaces)

Date: 23-10-2026

AIM

To configure a router by setting passwords and IP addresses on its interfaces, and to check the communication between two PCs in different subnets through the router.

SOFTWARE USED

Cisco Packet Tracer

THEORY

A router connects two or more separate networks (subnets) and routes data packets based on IP addresses. Each router interface acts as the **Default Gateway** for devices within its local network segment.

To prevent unauthorized administrative access, passwords are configured on the router: *Console password* secures direct terminal connections, *Enable password* protects Privileged EXEC mode, and *VTY passwords* secure remote Telnet/SSH management sessions. Router interfaces are disabled (shutdown) by default, requiring the `no shutdown` command for activation.

IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router (R1)	Gig0/0	192.168.1.1	255.255.255.0	N/A
Router (R1)	Gig0/1	192.168.2.1	255.255.255.0	N/A
PC2	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.2	255.255.255.0	192.168.2.1

PROCEDURE

1. Open Cisco Packet Tracer and place one 2911 Router and two PCs (PC2 and PC3).
2. Connect PC2 (NIC) to Gig0/0 of Router R1 and PC3 (NIC) to Gig0/1 using Copper Straight-Through cables.
3. Open Router R1 CLI tab. Type no to bypass System Configuration Dialog.
4. Set host name, console, enable, and VTY passwords using Cisco IOS commands.
5. Configure GigabitEthernet interfaces Gig0/0 and Gig0/1 with their designated IP addresses and enable them using `no shutdown`.
6. Configure PC2 with IP `192.168.1.2`, Subnet Mask `255.255.255.0`, and Default Gateway `192.168.1.1`.
7. Configure PC3 with IP `192.168.2.2`, Subnet Mask `255.255.255.0`, and Default Gateway `192.168.2.1`.
8. Open Command Prompt on PC2 and test connectivity using `ping 192.168.2.2`.

ROUTER IOS CONFIGURATION COMMANDS

1. Password Configuration

```
Router> enable Router# configure terminal Router(config)# hostname R1 R1(config)# enabl
```

TOPOLOGY & VERIFICATION

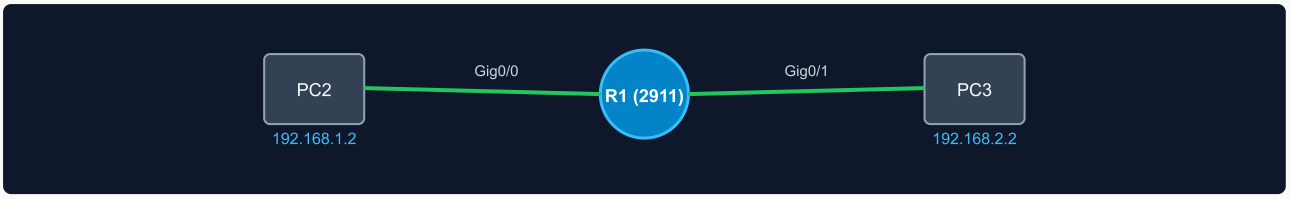


Fig 2.1 – Topology: PC2 and PC3 connected to 2911 Router

```
PC2> ping 192.168.2.2 Pinging 192.168.2.2 with 32 bytes of data: Request timed out. Reply from
```

Fig 2.2 – Command Prompt: Ping execution from PC2 to PC3 across subnets

Observation: The first request timed out while the router resolved ARP (Address Resolution Protocol) for the gateway interface. Subsequent packets received successful replies.

RESULT

Thus, the router was successfully configured with console/enable/VTY passwords and IP interfaces, establishing inter-subnet communication between PC2 and PC3.